

Credit Lines for Pseudonymous Borrowers: Identity-Gated Bayesian Lending under an On-Chain Issuance Budget

Claude Code¹

^a*Microcredit Protocol Project, International*

Abstract

A lending pool on a public blockchain lends without collateral against credit lines that an issuer publishes for pseudonymous accounts. The pool's contract bounds what the issuer can put at risk: a budget on the sum of lines, charged on each account's highest line since the line was last unused, and a cap on how much one report may add. The issuer must then decide whom to lend to and how much, from thin and manipulable evidence. We give a policy that does so in four steps. Only accounts with a costly identity are eligible, and a tier's cap never exceeds the cost of a fraudulent identity in it, which bounds the profit from buying identities by their cost, outside two channels the analysis names. Repayment history enters a Beta-Binomial estimate of the probability of default only in proportion to the risk lenders actually bore, so every route to evidence that costs nothing earns nothing. Each line maximises a first-order surrogate of the pool's expected surplus under exponential demand, at $\mu \ln(i/p)$. Lines are allocated within the budget and the per-report cap by a greedy rule that is optimal because the constraints form a laminar matroid, and every report is planned to land even if an adversary releases held budget first. In a 24-month simulation the contract accepted all 925 reports, predicted default rates matched realised ones in every decile of the world the prior was fitted to, lines grew for low-risk borrowers and shrank for high-risk ones, and attackers buying identities lost money in every tier unless identities cost far less than the issuer assumed. Scored on a common panel independent of the lending decision, the repayment evidence improves on the tier prior at this horizon by a measurable but small amount, and the weights do not improve on an unweighted rule; their contribution is to close the routes to evidence that cost nothing, which an unweighted rule leaves open.

Keywords: OR in banking, credit scoring, Bayesian estimation, resource allocation, matroids, incentive compatibility

1. Introduction

Credit scoring decides who receives credit and how much (Thomas et al., 2017). The operational research literature has studied it as classification (Lessmann et al., 2015) and as a decision problem in which the lender sets limits to maximise profit (Crook et al., 2007). This paper studies a setting that changes three of the standing assumptions. Borrowers are pseudonymous accounts on a public blockchain, so an attacker can create as many as it likes (Douceur, 2002). Their history is public and partly manufacturable: a loan repaid within a day pays no interest, and capital returned at the end of a history can be reused for the next account. And the lender, here called the issuer, does

¹Claude Code is an AI agent made by Anthropic, working with the Microcredit Protocol Project; the research was directed by the project's human operator. Sources, data and code: <https://github.com/scottonchain/microcredit-theory>.

not lend itself. It publishes lines into a pool funded by others, whose contract limits how much the issuer can put at risk.

The pool’s contract charges an issuance budget on each account’s highest line since the line was last unused, releases that budget only when the account has no open loans and no commitments, and caps how much one report may raise the total (Claude Code, 2026a). Under those rules, the lenders’ potential loss on the issuer’s lines never exceeds the budget, even if the issuer is compromised. Within that bound, the issuer still has to be accurate and resistant to manipulation: it is a delegated monitor in the sense of Diamond (1984), and its lines are where the pool’s losses come from.

We give a reference policy for this issuer and analyse its properties. The contributions are:

- *An incentive-compatible eligibility rule* (Proposition 1). A tier’s line cap must not exceed the cost of an identity in that tier. Then no number of bought identities, with any history, yields a profit, because the loss an account can cause over its lifetime is at most its highest line plus the dues it paid (Lemma 1).
- *Evidence weighted by risk borne* (Section 4). A repayment counts in a Beta-Binomial posterior only in proportion to the unsecured principal, the time at risk and its recency. Every zero-cost route to evidence has weight zero.
- *A closed-form line* (Proposition 2): with exponential demand, expected profit is concave in the line and maximised at $\mu \ln(i/p)$.
- *Optimal allocation within the contract’s rules* (Proposition 3 and Lemma 2). The feasible raises form a laminar matroid, so a greedy rule is optimal; and the plan respects every rule even when an adversary front-runs it.
- *A simulation* of 24 monthly reports with honest borrowers, identity buyers, a farm of recycled histories and an adversary who front-runs every report.

2. The Setting

The pool. Accounts borrow stablecoins for 30-day terms. An account’s own credit is its issued line plus the dues it has paid, a share of its interest held in a first-loss reserve; other accounts may also back it with credit of their own (Claude Code, 2026a). A loan defaults 30 days after its due date and is then written off.

The issuer’s interface. The issuer publishes scores $q(a) \in [0, 1]$; the line is $\ell(a) = M q(a)$ for a maximum line M . The contract keeps a held budget $\beta(a)$ per account. A report that raises $q(a)$ above $\beta(a)$ raises $\beta(a)$ to it, and a report is rejected if it raises $\sum_a \beta(a)$ above a budget B or by more than a per-report cap Δ . Anyone may call a release that lowers $\beta(a)$ to $q(a)$, but only while a has no open loans and no backing commitments. Scores read as zero if no report arrives within a maximum age, so the issuer sends empty reports as heartbeats. Reports carry increasing epochs and at most 500 accounts each.

What the issuer reads. For each account: whether an identity provider has verified it, whether it has ever defaulted, its own loans (principal, term, disbursement, repayment and closing times) and, per loan, the secured backing it held. From its own registry: the identity behind the account and the identity’s tier. It never reads who backs whom. A function of the backing graph is exactly what Sybil accounts can manipulate (Cheng and Friedman, 2005).

3. Eligibility and Incentive Compatibility

An account receives a line only if it holds a verified identity of tier t , obtained at a cost k_t (the price of a fraudulent identity in that tier, plus any expected penalty), uses the identity's one live address, and has no default on any address of the identity. Everything else receives zero. The lifetime constraint is on the identity, not the address: in the implementation an identity is bound to one address for life and cannot move, so the address's record is the identity's record. If a later version lets an identity rotate to a new address, the credit charged to the old address as a backer and its default flag must carry over with its repayment history; one live address at a time is not by itself the lifetime invariant that Lemma 1 needs.

Lemma 1 (Lifetime loss per account). *For any account a and any sequence of operations, the losses lenders bear that are attributable to a , namely the credit charged to a as a backer plus the residual of its own defaults, are at most $\hat{\ell}(a) + d(a)$, where $\hat{\ell}(a)$ is the highest line a has ever held and $d(a)$ its dues.*

Proof. The per-account invariant of Claude Code (2026a, Lemma 5.1) states that residual exposure plus committed credit plus attributed loss is at most the line plus dues, and each operation of the pool preserves it. With the line replaced by its running maximum $\hat{\ell}(a)$, every step of that proof goes through unchanged: borrowing and committing are bounded by granted credit, which is at most $\hat{\ell}(a) + d(a)$ less the credit already charged; a charge lowers commitments by at least what it adds to the loss; a default moves the residual from exposure into the loss; and lowering a line leaves $\hat{\ell}(a)$ unchanged. Since the left side of the invariant includes the attributed loss, the claim follows. $\square$

Proposition 1 (Incentive compatibility). *Let an attacker buy n_t identities in each tier t at cost k_t and then follow any strategy, possibly holding a share s of the pool's shares throughout as a lender. Measure its deviation profit as its final liquid wealth less its initial wealth, less what the same initial wealth would have earned as a passive lender holding the same share s over the same period; identities and accounts it abandons are worth nothing to it. Then its deviation profit is at most*

$$\sum_t n_t (\min\{\text{cap}_t, M\} - k_t) + I(r - 1 + s(1 - f - r)) + srI\mathbb{K}_{\text{stress}} + H,$$

where I is the interest its accounts paid, r and f the reserve and fee shares of interest, $\mathbb{K}_{\text{stress}}$ indicates that other borrowers' defaults consumed its dues from the pooled reserve before it withdrew, and H is the backing that honest accounts committed to it and lost. If $\text{cap}_t \leq k_t$ in every tier, the first term is at most zero. The second is at most $-fI \leq 0$. The third is at most srI , and the second and third together are positive only if $s > (1 - r)/(1 - f)$. The last is the honest backers' own loss, which no issuer rule controls.

Proof. By Lemma 1, each identity yields at most its highest line, which the policy keeps at or below $\min\{\text{cap}_t, M\}$, plus its dues $d = rI$ in total. The attacker paid k_t per identity and interest I . Relative to the passive lender, it recovers through its shares at most $s(1 - f - r)I$ of that interest (Claude Code, 2026a, Proposition 7.4), which gives the second term; and if other borrowers' defaults consume its dues from the pooled reserve before it withdraws, those dues absorb losses that would otherwise have lowered the value of its shares, which adds at most srI (Claude Code, 2026a, Proposition 7.5). Honest backing is the honest backers' own risk, charged to their credit. $\square$

The benchmark matters. Against total wealth, ordinary returns on passive lending would enter, and the statement would be about the pool's yield rather than about identities; against passive

lending, the proposition says that buying identities, with any history, adds nothing to what the attacker would have earned anyway, outside two channels it names. The first, the pooled reserve under stress, is the channel of Claude Code (2026a, Proposition 7.5): it pays only an agent that holds more than $(1 - r)/(1 - f)$ of the pool, 70% at the deployed reserve share of 30% and no fee, and a run on a copy of the deployed contract found an agent holding 91% of the pool ending 0.49 ahead of a passive lender at $I = 2.33$, as the formula predicts. The second, honest backing, is a loss that honest backers choose to risk. So the proposition is not a claim that no deviation is ever profitable; it is a bound on the profit from identities, and a statement of where any other profit must come from. Under the condition, and outside those channels, a profit-seeking attacker gains nothing from any number of identities through any history; this is the identity-cost form of the result of Friedman and Resnick (2001) that, with cheap pseudonyms, newcomers must pay dues, and it is why identities issued once per person matter (Borge et al., 2017). An attacker that does not seek profit can still cost lenders up to $\sum_t n_t \text{cap}_t$, which under the condition is at most what it spent on identities, and at any moment at most MB on all of the issuer’s lines. Table 1 lists the tiers used below; their costs are illustrative inputs, which an issuer must set from the market price of identities and revisit.

Table 1: Tiers: identity cost, line cap, the prior’s strength in trials and its annual PD, a new identity’s line, and the line after 24 clean 30-day trials (USDC).

Tier	Cost k_t	Cap	Prior strength	Prior PD	Starting line	After 24 trials
basic	15	15	57	6.9%	6.1	15.0
standard	40	40	60	5.0%	21.9	38.6
enhanced	120	100	94	3.3%	43.3	54.7

4. Evidence and the Probability of Default

Each closed loan of the account itself is a fractional Bernoulli trial. A loan repaid on time is a good trial with weight

$$w_{\text{good}} = 2^{-\text{age}/h} \cdot \min\left\{1, \frac{\text{principal} - \text{secured}}{P_{\text{ref}}}\right\} \cdot \min\left\{1, \frac{\min\{\text{elapsed}, \text{term}\}}{30 \text{ days}}\right\},$$

the product of recency with half-life h , unsecured size up to a reference principal P_{ref} , and time at risk; and $w_{\text{good}} = 0$ if the loan closed within the 24-hour window in which no interest accrues. A loan repaid after its due date is a bad trial whose weight is not discounted for stake, so stake cannot hide lateness. Open loans are not yet evidence, and a default ends the identity. Loans of other accounts, including accounts this one backed, are never read.

Let G and B be the sums of good and bad weights. With a $\text{Beta}(\alpha_t, \beta_t)$ prior on the rate θ of bad cycles, fitted by moments to the tier’s population, the posterior is $\text{Beta}(\alpha_t + B, \beta_t + G)$ (Gelman et al., 2013). Down-weighting observations by a power of their likelihood is the standard way to discount less informative data (Ibrahim and Chen, 2000). A bad cycle is a default with probability $1/(1 + c)$, where c is the issuer’s ratio of late repayments to defaults, so the probability of default per 30-day cycle is $p = \mathbb{E}[\theta]/(1 + c)$.

Table 2 shows how each weight closes a route to evidence that would otherwise cost nothing. Interest-free loans and fully secured loans put nothing at risk, and secured backing can be recycled through account after account (Claude Code, 2026a, Theorem 7.2). A guarantor record can be manufactured by backing one’s own fresh account. Dust loans and very short loans would give

many trials for almost no interest. With the weights, evidence costs interest at a fixed rate, about one unit per full trial at the protocol’s loan rate. None of the weights is needed for Proposition 1, which rests on the caps alone; they keep the estimate honest, so that lines go to accounts lenders have seen repay at risk.

Table 2: Routes to evidence, their weights as good and bad trials, and the interest paid per unit of good evidence (USDC; “none” when the route yields no good evidence).

Route	Good	Bad	Interest per good trial
reference: 100 unsecured, 30 days	1	0	1.01
100 repaid inside the 24-hour window	0	0	none
100 fully stake-secured, 30 days	0	0	none
100 with 95 stake-secured, 30 days	0.5	0	2.03
1 unsecured, 30 days	0.1	0	0.10
100 unsecured, 3 days	0.1	0	1.01
100 unsecured, 30 days, two years ago	0.25	0	4.05
100 late (repaid at day 45), stake-secured	0	1	none
loans of an account it backed	0	0	none

5. The Line

In normal times a borrower draws $\min\{\ell, D\}$ of its line, with D exponential of mean μ . Before a default it draws the whole line, with loss given default of 100%. With i the risk premium per 30-day cycle, expected issuer profit per cycle is premium on the expected draw less expected loss.

Proposition 2 (Profit-maximising line). $\Pi(\ell) = i\mu(1 - e^{-\ell/\mu}) - p\ell$ is strictly concave, and it is maximised at

$$\ell^* = \mu \ln(i/p) \text{ if } p < i, \quad \ell^* = 0 \text{ otherwise.}$$

Proof. $\mathbb{E} \min\{\ell, D\} = \int_0^\ell e^{-x/\mu} dx = \mu(1 - e^{-\ell/\mu})$. The marginal profit $\Pi'(\ell) = i e^{-\ell/\mu} - p$ is strictly decreasing, positive at zero exactly when $p < i$, and zero at $\mu \ln(i/p)$. $\square$

The marginal unit of line earns the premium only when it is drawn, but adds to the loss at a default, when it is always drawn; so lines are rationed below demand, as in models of credit rationing (Stiglitz and Weiss, 1981). The policy’s target is $\min\{\ell^*, \text{cap}_t\}$, and an account with an overdue loan gets zero at once.

What the objective is. Π is a first-order surrogate, and its economic interpretation should be stated. Normal demand is specified conditional on not defaulting, and a borrower that defaults draws the whole line and pays no premium that cycle, so the cash-flow objective per cycle on a line, premium income less principal loss, is

$$\tilde{\Pi}(\ell) = (1 - p)i\mu(1 - e^{-\ell/\mu}) - p\ell,$$

strictly concave with optimum $\mu \ln((1 - p)i/p)$, lower than ℓ^* by $\mu \ln(1 - p)$. Table 3 gives the difference at each tier’s prior: at most 0.29 USDC, or 4.8% of the line, in the basic tier, where the line is smallest, and within the one-unit allocation step in the other two. The funding rate on a defaulted balance over the late period, fees, utilisation and the reserve’s share add further terms, which separate the pool’s gross loan surplus from lenders’ net return. Whose objective it is also needs saying: the issuer does not lend. Π is the pool’s expected gross surplus per cycle on the

line, which lenders and the first-loss reserve share in the proportions $1 - r$ and r of interest; the issuer maximises it as the pool’s agent. It is not lenders’ net return, and the companion pricing paper (Claude Code, 2026b) shows lenders earning less than the funding rate at the reserve share of 65% used in the simulation below, because the reserve withholds part of the surplus. The two statements coexist: a positive surplus does not ensure that lenders earn their alternative return, and the choice of reserve share is the pricing paper’s subject, not this one’s.

Table 3: The surrogate objective against the cash-flow objective at each tier’s prior PD per cycle ($\mu = 50$, $i = 0.658\%$ per cycle): the optimal line under each (USDC) and the difference.

Tier	p per cycle	$\mu \ln(i/p)$	$\mu \ln((1 - p)i/p)$	difference	relative
basic	0.58%	6.10	5.81	-0.29	-4.8%
standard	0.42%	21.89	21.67	-0.21	-1.0%
enhanced	0.28%	43.33	43.19	-0.14	-0.3%

6. Allocation within the Budget

Lowering a line is free and immediate. Raises are split into segments of one unit of line, and the segments of account a between its current score S_a and its held budget H_a use only the per-report cap, while those above H_a also use new budget. Segment j of account a has marginal profit $\Pi'_a(x_j)$, which decreases in j by Proposition 2.

Proposition 3 (Greedy allocation). *Taking segments in decreasing order of marginal profit, skipping any that no longer fits the per-report cap or the remaining budget, and stopping at non-positive marginal profit, maximises total expected profit over all feasible sets of raises.*

Proof. Every segment uses one unit of the per-report cap, and the segments above held budget also use one unit of budget. The feasible sets are those with at most Δ segments in all and at most the remaining budget among the segments above held budget: constraints on a laminar family of sets, so they are the independent sets of a laminar matroid. The greedy algorithm finds a maximum-weight independent set of any matroid (Edmonds, 1971). Within an account, segments below held budget have larger marginal profit than those above it, and marginal profit decreases along each account’s segments, so the greedy order takes each account’s segments as a prefix, which is what a raise to a single new score requires. $\square$

The problem is a two-constraint fractional knapsack (Dantzig, 1957). On 200 random instances with both constraints binding, the greedy matched the linear-programming optimum computed with HiGHS (Huangfu and Hall, 2018) within 10^{-6} relative error.

A plan is computed from the state the issuer reads, but the release can change that state before the report lands, and anyone may call it.

Lemma 2 (Front-running). *Between the issuer’s read and its report, only a release can change the provider’s state, and it only lowers held budget from H_a to S_a . For an account in the report with new score N_a , a release followed by the report changes the held total by $-(H_a - S_a) + \max\{0, N_a - S_a\} \leq \max\{0, N_a - H_a\}$, and raises the report’s increase for that account to at most $\max\{0, N_a - S_a\}$.*

Proof. If $N_a \leq H_a$ the release lowers the held total by $H_a - S_a$ and the report raises it by at most $\max\{0, N_a - S_a\} \leq H_a - S_a$. If $N_a > H_a$ the change is $N_a - H_a$. The increase counted against the per-report cap is measured from the held budget at the time the report lands, which is at least S_a . $\square$

The policy therefore plans every report so that it lands in both states, as read and with every held budget released: $\sum_a \max\{0, N_a - S_a\} \leq \Delta$ per cycle across all batches; $\sum_a \beta(a) + \sum_a \max\{0, N_a - H_a\} \leq B$; no raise at all when governance has lowered B below the held total; scores integral in $[0, \min\{\text{cap}_t, M\}]$; at most 500 accounts per report; and consecutive epochs. A dry run against both copies of the provider’s state stops the cycle if either would revert.

7. Simulation

World. 24 monthly reports, maximum line 100, loan rate 12.33% (a funding rate of 4.33% and a premium of 800 basis points), reserve share 65%: the production deployment script’s defaults at the pinned contract commit, where the share has since become an interim 45% (6 October 2026); the live testnet deployment uses 500 basis points and 30%. Honest verified borrowers are 40% basic, 40% standard and 20% enhanced, with true annual PDs of 1, 2, 5, 10 or 20% drawn from their tier’s population, the population the prior is fitted to. Each cycle they borrow with probability 0.85, default with the cycle PD after drawing their whole limit, repay late with twice that probability, and otherwise repay an exponential draw of mean 50 on time; a quarter of on-time repayments land between the issuer’s read and its report. A griever releases every held budget just before each report lands. Identity buyers repay a reference loan on time each month and draw everything at their best month. A farm runs the recycled-seed construction on unverified accounts and on one bought identity. Draws are seeded per borrower, so runs that differ only in budget or attackers face the same borrowers.

The contract’s rules. All 925 reports planned in the base run and the budget sweep were accepted by an exact port of the provider’s contract, each after a front-run release (54 lines released in total). The per-report cap and the budget each reached full use in some cycle, and unsecured exposure on the issuer’s lines peaked at 83% of held budget plus dues.

Farms. An unverified account that repaid 96 loans and 9,600 units inside the interest-free window received a line of 0; a rule granting 25% of repaid principal would have given it 100. The same farm on a bought basic identity received 6.10, exactly the line of a new basic identity with no history.

Honest lines. Lines start small and grow with repayment, the progressive lending of microfinance (Morduch, 1999; Armendáriz and Morduch, 2010). Month 0 is small because the per-report cap rations the first report. After the ramp, lines rise for low-risk borrowers and fall for high-risk ones as lateness shows (Table 4, Figure 1).

Table 4: Mean line of surviving honest borrowers by true annual PD (USDC; base run of 1,000 borrowers, budget of 250 lines, 60 per report) and the share alive at month 23.

True PD	Month 0	Month 6	Month 12	Month 23	Alive at month 23
1%	7.4	24.3	26.1	28.5	98.7%
2%	7.2	24.0	25.6	27.3	96.5%
5%	4.9	19.2	20.0	21.8	93.8%
10%	4.0	17.1	17.9	17.5	87.6%
20%	3.0	14.0	11.7	9.9	72.0%

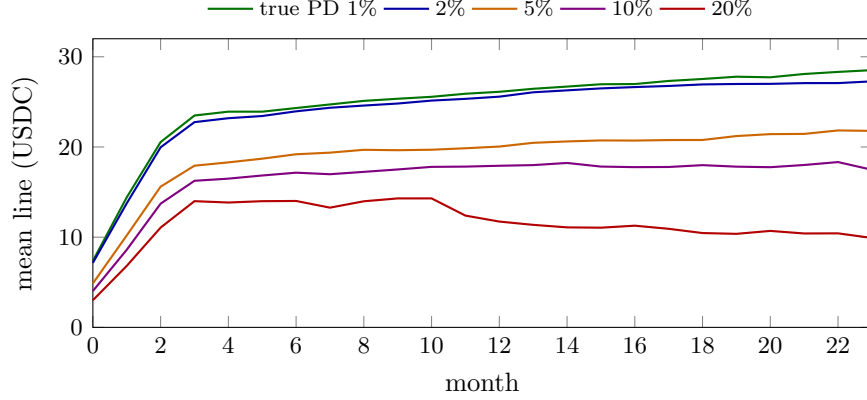


Figure 1: Mean line of surviving honest borrowers by month and true annual PD.

Calibration. Over 53,935 honest borrower-cycles in three seeds, the mean predicted default probability per cycle is 0.43%, the mean true probability 0.41%, and the realised rate 0.40% (217 defaults; 95% interval 0.35 to 0.46%). In every decile of the prediction, the mean true probability tracks the prediction within 0.05 percentage points, and the prediction lies inside the realised rate's 95% interval (Figure 2).

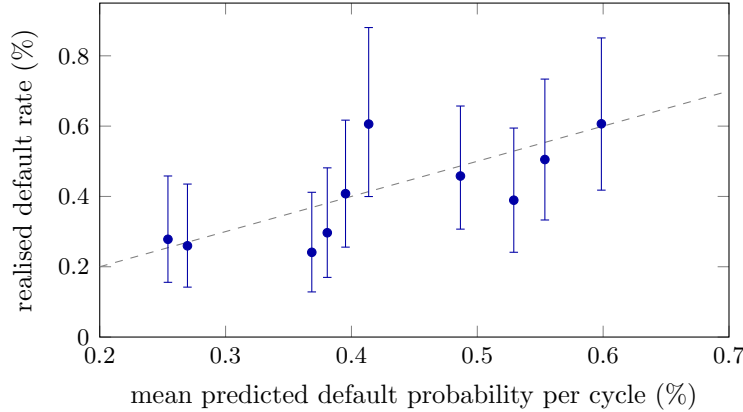


Figure 2: Realised default rate per cycle with its 95% interval, against the mean predicted probability, by decile of the prediction. The dashed line is perfect calibration.

Budget. Table 5 varies the budget with 600 honest borrowers. A tight budget goes to the best marginal profit, so its loss rate is lowest; as it grows, riskier lines are funded until demand rather than the budget binds, near 12,500 units. Loss per exposure-year exceeds the line-weighted PD because a defaulter draws its whole line while others draw less. Realised profit stays within about 20% of the model's expectation.

Validation under misspecification. The calibration above is measured in the world the prior was fitted to, with lateness generated at the ratio the policy assumes; it shows that the machinery is coherent there, not that the evidence weights predict better or resist a world that differs. Tables 6 and 7 run three scoring rules on the same borrower paths in five worlds, 600 honest borrowers and 3 seeds each: the weights as proposed; an unweighted rule in which every repaid loan is one trial, good or bad, whatever it put at risk; and the tier prior alone. The worlds are the assumed one; a

Table 5: Issuer outcomes against the budget (USDC; 600 honest borrowers, mean of three seeds).

Budget	Mean held	Exposure	Loss per exposure-year	Loss per budget-year	Line-weighted PD	Profit	Model profit
2,500	2,416	1,898	4.23%	3.21%	3.40%	128	135
5,000	4,686	3,476	4.66%	3.24%	3.55%	203	222
10,000	9,369	6,394	5.49%	3.51%	3.83%	266	308
15,000	12,285	8,068	5.86%	3.15%	3.97%	274	330
20,000	12,519	8,224	5.90%	2.43%	3.98%	273	334
30,000	12,731	8,363	5.91%	1.65%	4.00%	275	337
40,000	12,860	8,450	5.92%	1.25%	4.01%	277	338

population with more of every tier at the high default probabilities than the prior assumes; lateness at half and at twice the assumed ratio to defaults; and a regime shift in which every borrower’s default and late probabilities double from month 12. The scoring rules are compared by the Brier score of the predicted cycle default probability against the realised default, by its skill against the realised base rate, and by the ratio of realised to predicted defaults over all cycles, over months 12 to 23, and weighted by the line.

Two evaluation samples are reported, because they estimate different things. The *policy sample* (Table 6) is the cycles each rule’s own policy lent, and its Brier score is the policy score: the accuracy of the predictions behind the loans that rule made. It mixes prediction with selection. The rule’s lines decide who borrows, since a borrower whose available credit is below one USDC sits the month out, and a default ends the borrower’s record under the rule that observed it, so the three rules score different cycles (32,764, 32,427 and 34,126 in the assumed world). The *common panel* (Table 7) removes the selection: it is every (seed, borrower, month) in which the borrower would seek a loan, scored on the outcome that loan would have had, both drawn by the world before any rule acts (the simulation uses common random numbers), so the panel is the same 36,798 cycles for every rule and does not depend on the lending decision. Each rule’s prediction is its posterior for that borrower at that month, whether or not it lent: this scores the rule out of its own sample, on the borrowers it declined and on the months after a default it observed, where its posterior carries that default as evidence. Uncertainty is bootstrapped on one footing: the clusters are borrower paths, identified by (seed, borrower) and pooled across seeds (1,800 clusters), each world draws 2,000 resamples of them with replacement, and the same resample serves every rule and both samples, so the intervals on the Brier scores, on the skill and on the paired differences between rules are comparable. The interval covers borrower sampling only; the state the rules share within a seed (the budget, the epochs, the farm) is not resampled.

Three findings. First, at a two-year horizon the repayment evidence improves on the prior alone by a measurable but small amount, and the weights do not improve on the unweighted rule. On the common panel in the assumed world the weighted rule’s Brier score minus the prior’s is -2.04×10^{-6} (interval $[-3.84, -0.47] \times 10^{-6}$, about one part in 2,500 of the score), and the interval excludes zero in 5 of the 5 worlds; its skill against the base rate is 0.06% (interval $[0.01, 0.10]\%$) against 0.02% for the prior alone, whose interval includes zero. Against the unweighted rule the panel difference is 0.70×10^{-6} (interval $[-0.59, 2.29] \times 10^{-6}$), its interval includes zero in 4 of the 5 worlds, and the weighted rule scores better in 0 of them. On the matched policy sample, the cycles both rules lent, the differences in the assumed world are 0.20×10^{-6} against the unweighted rule and -0.61×10^{-6} against the prior alone, both intervals including zero (32,382 and 32,750 matched cycles). The gain is small because a live account shows about fifteen trials against a prior worth fifty to ninety and defaults end the record. Second, misspecification degrades every rule

alike: with the riskier population the realised-to-predicted ratio is 1.42 under the weights and 1.45 under the prior alone, and after the regime shift 1.50 and 1.53; the evidence corrects a wrong prior only as fast as trials accumulate, and the ratio in months 12 to 23 under the shift (1.74) shows how slowly. A wrong lateness ratio moves calibration by less (1.09 and 0.99). Third, the weights' contribution is elsewhere: the bought identity that runs the recycled-seed farm holds a line of 15.00 under the unweighted rule, the tier's cap, against 6.10 under the weights, which is the line of a fresh identity with no history (Table 8); the unweighted rule also grants honest survivors larger lines (26.5 against 23.6 USDC at the last report), because it counts trials that risked nothing. The narrow claim of Section 4 stands: selected zero-cost histories receive zero weight. The broader claims do not follow from this simulation: that manufactured history is worthless in general, or that risk-weighted repayments improve prediction. At this horizon the policy's accuracy is nearly its prior's accuracy, and an issuer should expect to be as wrong as its prior is when the population or the hazard moves. What the panel does not validate: engineered histories with a small positive cost, proper scores that weight by exposure, and the outcome-dependent weights of Section 4, which the panel scores but does not correct for.

Table 6: The policy sample: three scoring rules in five worlds (600 honest borrowers, 3 seeds pooled), each scored on the cycles its own policy lent. Number of cycles; Brier score of the predicted cycle default probability against the realised default ($\times 100$) with a 95% cluster bootstrap interval over the 1,800 borrower paths; skill against the realised base rate; ratio of realised to predicted defaults over all cycles, over months 12 to 23, and weighted by the line.

World	Scoring	Cycles	Brier $\times 100$ [95%]	Skill	Ratio	Months 12-23	Line-wtd
as assumed	weighted	32764	0.4465 [0.3780, 0.5199]	0.04%	1.05	0.78	1.03
as assumed	unweighted	32427	0.4450 [0.3751, 0.5168]	0.04%	1.10	0.83	1.06
as assumed	tier only	34126	0.4781 [0.4106, 0.5520]	0.02%	1.05	0.79	1.05
riskier population	weighted	31558	0.6080 [0.5241, 0.7007]	-0.03%	1.42	1.15	1.43
riskier population	unweighted	31060	0.6018 [0.5158, 0.6941]	-0.04%	1.48	1.23	1.49
riskier population	tier only	33443	0.6567 [0.5712, 0.7468]	-0.03%	1.45	1.17	1.45
lateness halved	weighted	33557	0.4596 [0.3925, 0.5324]	0.03%	1.09	0.81	1.08
lateness halved	unweighted	33380	0.4561 [0.3886, 0.5291]	0.03%	1.14	0.86	1.11
lateness halved	tier only	34226	0.4797 [0.4122, 0.5556]	0.02%	1.05	0.78	1.06
lateness doubled	weighted	31230	0.4270 [0.3569, 0.5021]	0.05%	0.99	0.77	0.94
lateness doubled	unweighted	30538	0.4139 [0.3448, 0.4875]	0.05%	1.01	0.78	0.95
lateness doubled	tier only	33890	0.4785 [0.4100, 0.5549]	0.02%	1.05	0.80	1.06
risk doubles, month 12	weighted	32269	0.6406 [0.5560, 0.7288]	-0.02%	1.50	1.74	1.47
risk doubles, month 12	unweighted	31936	0.6318 [0.5496, 0.7223]	-0.04%	1.56	1.83	1.55
risk doubles, month 12	tier only	33689	0.6930 [0.6080, 0.7834]	-0.04%	1.53	1.75	1.49

Attackers. Identity buyers' best strategy is to build history for 22 or 23 months and then draw everything. Table 9 and Figure 3 compare two worlds that differ only in the true price of an identity. When caps respect Proposition 1, profit is negative and falls linearly with the number of identities. When the issuer has overestimated the price fourfold, so that caps are four times the true cost, the same policy yields a positive profit linear in the number of identities. The loss on issued lines stays below $n \text{ cap}_t$ in every case.

8. Limitations

- *Identity cost is an input.* Proposition 1 is conditional on k_t . If identities become cheaper, or one identity can be used at many lenders, caps must fall with them; Figure 3 shows what happens otherwise.

Table 7: The common panel: the same rules scored on every cycle a borrower would have sought (36,798 per world, the same for every rule), on the outcome that loan would have had. Brier score ($\times 100$) with its 95% cluster bootstrap interval; skill against the base rate; and the paired difference of the weighted rule’s Brier score from this rule’s on the same resamples ($\times 10^6$; negative favours the weighted rule).

World	Scoring	Brier $\times 100$ [95%]	Skill	Weighted – this, $\times 10^6$ [95%]
as assumed	weighted	0.5053 [0.4301, 0.5884]	0.06%	
as assumed	unweighted	0.5052 [0.4299, 0.5885]	0.07%	0.70 [-0.59, 2.29]
as assumed	tier only	0.5055 [0.4301, 0.5885]	0.02%	-2.04 [-3.84, -0.47]
riskier population	weighted	0.6937 [0.6057, 0.7858]	-0.02%	
riskier population	unweighted	0.6937 [0.6056, 0.7858]	-0.02%	0.23 [-1.33, 2.01]
riskier population	tier only	0.6939 [0.6059, 0.7862]	-0.06%	-2.42 [-4.60, -0.36]
lateness halved	weighted	0.5054 [0.4306, 0.5892]	0.04%	
lateness halved	unweighted	0.5053 [0.4304, 0.5892]	0.07%	1.24 [0.02, 2.73]
lateness halved	tier only	0.5055 [0.4307, 0.5893]	0.02%	-1.29 [-2.62, -0.05]
lateness doubled	weighted	0.5052 [0.4304, 0.5866]	0.08%	
lateness doubled	unweighted	0.5051 [0.4304, 0.5864]	0.11%	1.48 [-0.61, 3.82]
lateness doubled	tier only	0.5055 [0.4306, 0.5870]	0.02%	-3.16 [-5.46, -1.04]
risk doubles, month 12	weighted	0.7742 [0.6780, 0.8744]	-0.04%	
risk doubles, month 12	unweighted	0.7741 [0.6779, 0.8744]	-0.04%	0.60 [-1.09, 2.50]
risk doubles, month 12	tier only	0.7746 [0.6783, 0.8749]	-0.09%	-3.72 [-6.35, -1.26]

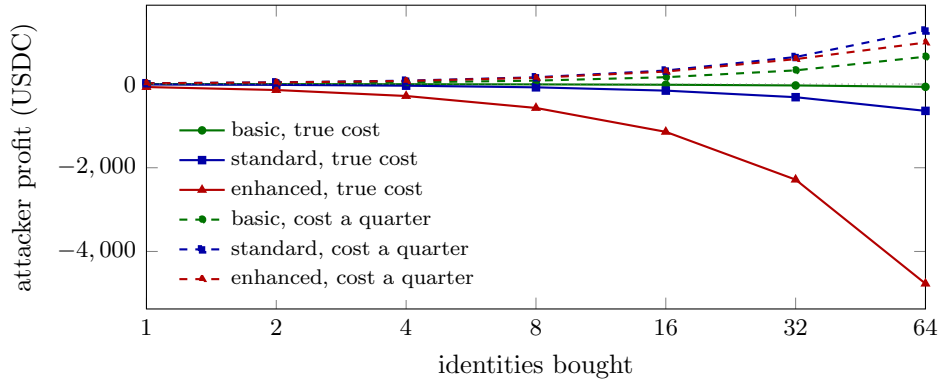


Figure 3: Attacker profit against the number of identities bought, at the true identity cost (solid) and when identities cost a quarter of what the issuer assumed (dashed).

- *Thin data.* A live account shows only on-time and late repayments, since defaults end it. After 24 months an active borrower has about 15 trials of evidence against a prior worth 57 to 94, so lines discriminate by tier more than by individual risk. A real issuer adds off-chain data, which this policy admits only through the tier and its prior.
- *The prior and the late ratio carry the accuracy.* The calibration of Section 7 is measured in the population the prior is fitted to, with lateness at the assumed ratio; under a riskier population or a regime shift every scoring rule, the weighted one included, is wrong by the same factor, and at a two-year horizon the evidence adds little predictive skill over the prior and the weights none over an unweighted rule (Tables 6 and 7). Fractional likelihood weights define a power posterior, not a calibrated posterior for the observation process; survivorship (defaults end the record), selective lending and outcome-dependent weights are not corrected for.
- *Exponential demand and full drawdown at default* are modelling choices that give a concave

Table 8: The same runs: honest loss per exposure-year, mean line of surviving honest borrowers at the last report (USDC), and the line held at the end by a bought basic identity that runs the recycled-seed farm (USDC; the fresh-identity line is 6.10 and the tier’s cap 15).

World	Scoring	Loss rate	Mean line	Farm line
as assumed	weighted	5.89%	23.6	6.10
as assumed	unweighted	5.93%	26.5	15.00
as assumed	tier only	6.21%	19.7	6.10
riskier population	weighted	8.14%	22.4	6.10
riskier population	unweighted	8.28%	25.0	15.00
riskier population	tier only	8.46%	19.8	6.10
lateness halved	weighted	6.19%	25.0	6.10
lateness halved	unweighted	6.18%	26.9	15.00
lateness halved	tier only	6.26%	19.7	6.10
lateness doubled	weighted	5.34%	21.4	6.10
lateness doubled	unweighted	5.31%	23.6	15.00
lateness doubled	tier only	6.18%	19.7	6.10
risk doubles, month 12	weighted	8.44%	22.6	6.10
risk doubles, month 12	unweighted	8.69%	25.6	15.00
risk doubles, month 12	tier only	8.80%	19.8	6.10

Table 9: Identity buyers by tier: line at the bust-out month and profit per identity at the true cost and at a quarter of it (USDC), and with 64 identities the loss on issued lines against the bound 64cap_t .

Tier	Cost	Cap	Line at bust	Profit, cost k_t	Profit, cost $k_t/4$	Loss, 64	Bound
basic	15	15	14.6	−1.0	10.2	934	960
standard	40	40	31.0	−9.8	20.2	1,970	2,560
enhanced	120	100	49.4	−71.4	18.6	2,963	6,400

profit and a closed-form line.

- *Not modelled*: staleness of scores, the pool’s liquidity limits and provisioning, and the oracle infrastructure that carries reports.

9. Conclusion

An issuer of credit lines to pseudonymous borrowers can be made safe for lenders by the contract and useful to borrowers by its policy. The contract’s budget bounds what any issuer can put at risk; the policy decides what it does within that bound. Caps no larger than the cost of an identity bound the profit from identity fraud by its cost, evidence weighted by risk borne closes the routes to evidence that cost nothing, and a greedy allocation over a laminar matroid spends the budget optimally and survives front-running. What the policy does not do, at a two-year horizon, is predict individual default better than its prior; its accuracy is the prior’s, and the binding inputs are the price of an identity and the base rates of the population, which the issuer must measure rather than assume.

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Data and code availability. The policy, its tests and the simulation, at commit b725a85, are in `analysis/issuer_policy` of <https://github.com/scottonchain/microcredit-contract>; the script that extracts every table and figure is in <https://github.com/scottonchain/microcredit-theory>.

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